

Graph-Aware Anomaly Detection for Blockchain Forensics: A 48-Experiment Benchmark on the Elliptic Bitcoin Dataset

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Abstract—Detecting illicit activity on a public blockchain involves facing three concurrent challenges: severe class imbalance (2 % illicit among all transactions), temporal concept drift (illicit rate shifting from 11.6 % in training to 6.5 % at test time), and a pseudonymous graph whose nodes carry 165 anonymous features. We present a systematic benchmark of 48 model variants across five paradigms—unsupervised, supervised, deep neural, graph neural network (GNN), and semi-supervised—on the Elliptic Bitcoin dataset (203 769 transactions, 234 355 directed edges), organized into five iterative improvement rounds. All models share a strict temporal train/test split that respects the time-series nature of the data. The best result, $F1 = 0.8265$, is achieved by Gradient Boosting augmented with seven structural topology features. GraphSAGEv2 reaches $F1 = 0.698$ using graph message passing over node features and topology, recovering 84.5 % of the supervised ceiling. Classical feature-space unsupervised baselines never exceed $F1 = 0.076$; the course-derived DOMINANT graph autoencoder reaches $F1 = 0.212$. A controlled GNN ablation shows that LayerNorm and self-loops (+0.068 $F1$) matter more than depth or attention. β -VAE with $\beta=0.01$ achieves $F1 = 0.407$, the best purely unsupervised result.

Index Terms—anomaly detection, blockchain forensics, GraphSAGE, DOMINANT, gradient boosting, β -VAE, EVT threshold, SHAP, Optuna, class imbalance, Elliptic dataset

I. INTRODUCTION

Bitcoin’s blockchain is fully public yet pseudonymous: every transaction is permanently recorded, but the identities behind wallet addresses remain hidden. This pseudonymity has made Bitcoin a vehicle for money laundering, ransomware payments, and darknet market settlements. Traditional anti-money-laundering (AML) controls rely on verified identity, while blockchain forensics must infer intent from transaction patterns and graph structure.

The Elliptic dataset [1] is the largest publicly available labeled graph of real Bitcoin transactions: 203 769 transactions, 234 355 directed edges, 4 545 labeled illicit (2 %), 42 019 labeled licit (21 %), and 157 205 unlabeled (77 %) over 49 time steps spanning roughly two years. Each node carries 165 anonymous features, 93 local statistics and 71 aggregated neighborhood summaries (some sources count 166 including

the time-step index, which we treat as a split attribute rather than a predictive feature).

Three intertwined challenges:

- 1) **Severe imbalance.** 9.76 % illicit among labeled nodes; 6.5 % in the test window. A naive majority classifier achieves 93.5 % accuracy while detecting zero illicit transactions, making accuracy a completely uninformative metric.
- 2) **Temporal concept drift.** Illicit rate drops from 11.6 % to 6.5 % from train to test (44 % relative decline). Random shuffling leaks future patterns into training, inflating every metric.
- 3) **Anonymous features.** Elliptic has not published feature semantics, so cross-model feature agreement is the only validation of signal robustness.

A. Research Questions

- **RQ1:** Do classical unsupervised detectors provide a viable AML baseline, or do illicit transactions fail to form anomalous clusters?
- **RQ2:** How much does graph structure add over tabular supervised models, and what fraction of the supervised ceiling can a GNN recover?
- **RQ3:** Which GNN design choices (depth, width, normalization, attention) matter most on a sparse graph (avg. degree 2.3)?
- **RQ4:** Do any features rank consistently across heterogeneous model families?

B. Contributions

- 1) Strict **temporal evaluation protocol** (steps 1–34 train / 30–34 validation / 35–49 test) applied to all 48 variants.
- 2) **48-variant benchmark** across five improvement rounds covering classical anomaly detection, supervised trees, deep autoencoders, GNNs, semi-supervised pseudo-labels, and topology-augmented methods.
- 3) **10-experiment GNN ablation** isolating hidden dimension, depth, LayerNorm, self-loops, and attention heads.

- 4) **Optuna TPE tuning** [14] (40 trials each) for GBM, RF, and GraphSAGE.
- 5) Evaluation of **course-derived methods**: DOMINANT [6], LSTM-AE, β -VAE [7], spectral embedding, EVT thresholding [15], and pseudo-label propagation [16].
- 6) **Cross-paradigm explainability** via SHAP and gradient attribution.

II. RELATED WORK

Weber et al. [1] introduced the Elliptic dataset and benchmarked LR, RF, and GCN, but did not explore unsupervised methods, deeper GNN architectures, or the score-inversion phenomenon in reconstruction-based detectors.

GNNs for fraud. Hamilton et al. [3] introduced GraphSAGE; Velickovic et al. [4] proposed GAT; Brody et al. [5] fixed GAT’s expressivity with GATv2. Pareja et al. [2] extended GCN to dynamic graphs (EvolveGCN). Alarab et al. [17] report an augmented GCN at F1 = 0.74 on Elliptic; reported EvolveGCN and self-supervised GraphSAGE results reach F1 = 0.75–0.77 (see Table XIII and §IV-J).

Graph-leakage critique. Maganti [18] re-evaluates GNNs on Elliptic under a *strict-inductive* protocol (training message passing restricted to steps ≤ 34 , mirroring deployment) versus the *transductive* protocol used by essentially all prior Elliptic papers, where the encoder’s forward pass sees the full graph – including test-period (steps 35–49) edges and nodes – at every training step, and only the loss is masked to labeled train nodes. Matched-seed GraphSAGE scores F1 = 0.294 transductive vs. F1 = 0.689 strict-inductive, a 39.5-point gap ($p=2.6 \times 10^{-12}$); under the strict-inductive protocol, Random Forest on raw features (F1 = 0.821) beats every GNN tested. §IV-J discusses this critique in the context of our own GNN training protocol.

Graph anomaly detection. Ding et al. [6] proposed DOMINANT, a GCN autoencoder that jointly reconstructs node attributes and graph structure without labels.

Reconstruction-based detection. Standard autoencoders [8] flag high reconstruction error as anomalous. On Elliptic we observe inversion: illicit transactions reconstruct with *lower* MSE [9]. Denoising AE [10] and VAE [11] are evaluated.

β -VAE [7] adds weight β to the KL term in the ELBO:

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - \beta D_{\text{KL}}(q_\phi(z|x) \| p(z)). \quad (1)$$

Lower β is reconstruction-dominated; $\beta=0.01$ yields best F1 (0.407) since reconstruction dominance preserves the inversion signal.

EVT thresholding. The Generalized Pareto Distribution (GPD) models score tails to set thresholds without labels [15]:

$$t_{\text{EVT}} = u + \frac{\hat{\sigma}}{\hat{\xi}} \left(\left(\frac{np}{N_u} \right)^{-\hat{\xi}} - 1 \right), \quad (2)$$

where u is the tail cutoff, N_u exceedances, and p exceedance probability.

TABLE I: Elliptic Dataset Statistics

Property	Value	Note
Transactions	203 769	total nodes
Directed edges	234 355	avg. degree 2.30
Illicit (label 1)	4 545	2 % of total
Licit (label 2)	42 019	21 % of total
Unlabeled	157 205	77 % of total
Time steps	49	steps 44–49 unlabeled
Local features	93	1 f_1–1 f_93
Aggregated feat.	71	a f_1–a f_71

III. DATASET AND METHODOLOGY

A. Dataset

The Elliptic dataset was published by Elliptic in collaboration with MIT. Feature semantics are not published; columns carry only numerical indices.

B. Temporal Split and Evaluation Protocol

A strict temporal split is mandatory: Bitcoin transactions have causal ordering, and random shuffling inflates all metrics by allowing interpolation between past and future behavior.

TABLE II: Temporal Data Split (by time step index, 1–49)

Partition	Labeled rows	Illicit %
Train (steps 1–34)	29 894	11.6 %
Validation (steps 30–34)	subset of train	—
Test (steps 35–49)	16 670	6.5 %
Unsup. train (steps 1–34)	136 265 all rows	—
Licit-only (steps 1–34)	26 432	0 %

The 44 % relative drop in illicit rate confirms genuine concept drift. **Primary metric:** F1 on the illicit class. Secondary: PR-AUC (threshold-independent, informative under imbalance) and ROC-AUC.

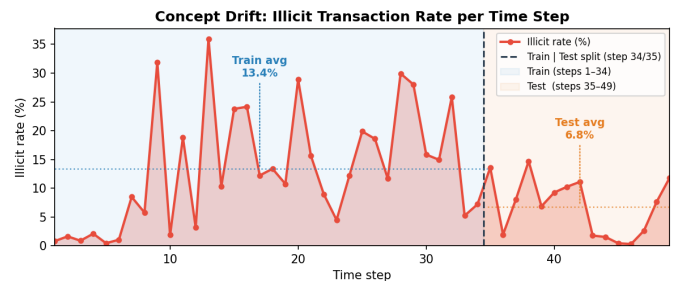


Fig. 1: Illicit transaction rate per time step (labeled nodes only). The rate falls from roughly 11.6 % in the training window (steps 1–34) to 6.5 % in the test window (steps 35–49), a 44 % relative decline confirming genuine concept drift. Random shuffling would mask this shift and inflate all metrics.

C. Per-Paradigm Training Setup

Each paradigm uses different training data and threshold strategy, as summarised in Table III. *Note on supervision:*

models in the IsoForest/LOF and OCSVM/AE/VAE rows use no labels for *training*, but their classification thresholds are selected by optimizing F1 on a labeled subset. They are therefore better described as *semi-supervised at threshold-selection stage* rather than fully unsupervised; truly label-free thresholds (EVT) are evaluated separately.

TABLE III: Per-Paradigm Training and Threshold Strategy

Paradigm	Training data	Threshold
IsoForest / LOF	136 265 all train rows	F1-opt, labeled train
OCSVM / AE / VAE	26 432 licit-only rows	F1-opt, val steps 30–34
Supervised	29 894 labeled train	0.5 (default)
GNN	Full graph, train masks	0.5 (default)
Ensemble	Labeled train rows	Grid search, val steps
Semi-sup.	Train + pseudo-labeled	0.5 (default)

D. Model Families

1) *Classical Unsupervised*: Isolation Forest [12] (contamination=0.1); LOF (transductive, applied to test set directly); One-Class SVM (RBF, $\nu=0.05$, licit-only train). Also evaluated: IsoForest on 7 structural topology features, and IsoForest/OCSVM on the top-50 spectral eigenvectors of $\tilde{A} = D^{-1/2}AD^{-1/2}$.

2) *Supervised Tree Models*: All use `class_weight='balanced'` or `scale_pos_weight`: Logistic Regression (L2), Random Forest (100 trees), Gradient Boosting (Optuna-tuned: $n=245$, $\text{depth}=8$, $\text{lr}=0.121$, $\text{sub}=0.918$), LightGBM (500 trees, $\text{max_depth}=8$), RF + graph degree/PageRank features, GBM + 7 structural topology features (degree, clustering, ego-density, OddBall), GBM + 330 temporal rolling features (5-step window), and GBM retrained on pseudo-labeled unlabeled nodes.

3) *Neural Anomaly Detection*: All trained on licit-only data; anomaly score = $-\text{MSE}$ (direction auto-detected on validation via AUC comparison): AE ($166 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow \dots \rightarrow 166$), DenoisingAE (latent=8, $\sigma_{\text{noise}}=0.1$), VAE and β -VAE grid search ($\beta \in \{0.01, 1, 2, 4, 8\}$) [7], [11], LSTM-AE (per-timestep mean features, $T=10$ sliding windows), and DOMINANT (2-layer GCN encoder, joint attribute+structure reconstruction) [6].

4) *GNN (Supervised)*: Full PyG graph; edges made undirected; inverse-frequency `CrossEntropyLoss`. GraphSAGE mean aggregation at layer k :

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{MEAN} \left(\{ h_v^{(k-1)} \} \cup \{ h_u^{(k-1)} : u \in \mathcal{N}(v) \} \right) \right). \quad (3)$$

Variants: baseline (2L, $h=128$, 200ep), GraphSAGEv2 (3L, $h=256$, LayerNorm, self-loops, 250ep), Optuna-tuned ($\text{lr}=0.0038$, $\text{dropout}=0.45$, 274ep), SAGEv2 + pseudo-labels (130 655 nodes), GAT (2L, 4 heads), and GATv2 (3L, 2 heads, residual, LayerNorm).

5) *Ensemble Methods*: Soft-vote ensemble of RF+LGB+GAT (threshold on train, Round 1) and GBM+LGBM+SAGEv2 (threshold grid-searched on validation). Four weight schemes tested; equal weights (1/3 each) won.

E. EVT Thresholding

GPD fitted to the upper tail of reconstruction scores (tail quantile $q=0.90$, exceedance probability $p=0.065$) per Eq. 2. Evaluated on all AE-based models.

F. Implementation

NVIDIA RTX 5070 Ti (12 GB VRAM, CUDA 12.8); Python 3.12, PyTorch 2.11, PyG 2.8, scikit-learn 1.9, LightGBM 4.6, SHAP 0.52, Optuna 4.9, NetworkX 3.6. `StandardScaler` fitted on train only; random seed 42 throughout.

IV. EXPERIMENTS

Experiments follow a logical progression: (1) establish unsupervised floor (RQ1); (2) quantify supervised ceiling and tuning gains; (3) diagnose neural reconstruction ceiling; (4) systematic GNN ablation (RQ3); (5) course-derived advanced methods; (6) cross-paradigm explainability (RQ4). All models are evaluated on the same 16 670 labeled test rows with no test-set leakage.

A. Unsupervised Baselines

TABLE IV: Classical Unsupervised Anomaly Detection

Model	F1	ROC-AUC	PR-AUC
Isolation Forest	0.021	0.172	0.036
One-Class SVM	0.023	0.235	0.039
LOF	0.076	0.506	0.066
IsoForest (structural)	0.094	0.458	0.056
IsoForest (spectral K=50)	0.056	0.407	0.052
OCSVM (spectral K=50)	0.061	0.391	0.054

All detectors score below F1 = 0.10. LOF’s AUC of 0.506 is random. Spectral embedding fails because the Fiedler value ≈ 2.0 (near bipartite maximum) indicates no sparse cut separating illicit nodes—money laundering chains are embedded within the spectral normal subspace.

Answer to RQ1: illicit transactions do *not* form anomalous clusters in feature space. Automated laundering produces stereotyped, regular transactions indistinguishable from high-frequency legitimate activity by density-based methods.

B. Supervised Models

TABLE V: Supervised Model Results (sorted by F1)

Model	F1	ROC-AUC	PR-AUC
Logistic Regression	0.302	0.881	0.291
GBM (baseline)	0.766	0.914	0.785
RandomForest (baseline)	0.801	0.935	0.794
RF + Graph Features	0.807	0.942	0.800
LightGBM	0.817	0.930	0.804
GBM + Temporal Rolling	0.818	0.930	0.802
GBM (Optuna tuned)	0.824	0.920	—
GBM + PseudoLabels	0.824	0.924	0.804
GBM + Structural	0.827	0.924	0.803

Logistic Regression achieves recall = 0.87 but precision = 0.18 (F1 = 0.302)—unusable without heavy post-filtering. The jump from LR to RF ($\Delta = +0.499$) reflects non-linear feature interactions that linear models cannot capture. Adding graph-topology features to RF gives only $\Delta = +0.006$ because the 71 aggregated features already encode neighborhood information in tabular form.

GBM Optuna reaches F1 = 0.824 ($\Delta = +0.058$ over baseline); the learning-rate–depth–subsample interaction is non-obvious and benefits from systematic search. RF Optuna ($\Delta = -0.004$) confirms RF is near its ceiling.

GBM with 7 structural topology features reaches **F1 = 0.8265**, the overall best result ($\Delta = +0.002$ over Optuna GBM); topology information is not fully subsumed by the 71 aggregated features.

GBM with 330 temporal rolling features (5-step window) improves AUC (0.920 $\rightarrow$ 0.930) but reduces F1 (0.824 $\rightarrow$ 0.818); global temporal context adds noise to per-transaction discrimination.

Semi-supervised GBM with pseudo-labels reaches F1 = 0.8235 ($\Delta = -0.0006$); GBM has already learned the optimal boundary from 29 894 labeled rows.

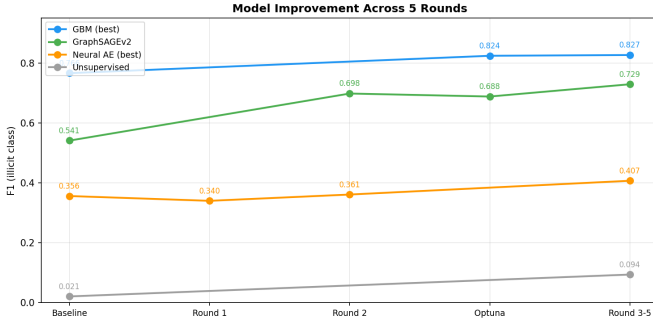


Fig. 2: Improvement across five rounds for key model families. “GBM (best)” tracks the highest-scoring GBM variant at each stage; “Neural AE (best)” tracks the highest-scoring autoencoder variant. Both labels include “(best)” to indicate they represent the family ceiling, not a single fixed model. GBM peaks at F1 = 0.827; GraphSAGEv2 reaches 0.729; the autoencoder family peaks at 0.407 (β -VAE, $\beta=0.01$).

C. Neural Anomaly Detection

1) *Score Inversion*: The central finding is **score inversion**: illicit transactions reconstruct with *lower* MSE than licit ones, so the anomaly score must be $-MSE$. Automated laundering produces stereotyped, templated patterns that are easier to reconstruct than diverse legitimate activity. Direction is auto-detected on validation by comparing $AUC(y, MSE)$ vs. $AUC(y, -MSE)$.

2) β -VAE Analysis: At $\beta=0.01$ (reconstruction-dominated), F1 = 0.407—the best unsupervised non-graph result. As β increases above 1, the KL term compresses licit and illicit representations together, weakening the anomaly signal. The F1 plateau at ≈ 0.36 –0.41 reflects the hard ceiling of unsupervised reconstruction-based detection on this dataset.

TABLE VI: Neural Anomaly Detection Results (sorted by F1)

Model	F1	ROC-AUC	PR-AUC
LSTM-AE	0.122	0.446	0.064
VAEv2	0.256	0.732	0.161
β -VAE ($\beta=2.0$)	0.280	0.746	0.211
β -VAE ($\beta=4.0$)	0.313	0.754	0.287
VAE ($\beta=1.0$)	0.340	0.778	0.310
β -VAE ($\beta=8.0$)	0.350	0.763	0.299
AE (baseline)	0.356	0.781	0.336
DenoisingAE	0.361	0.792	0.366
β -VAE ($\beta=1.0$)	0.372	0.776	0.338
β-VAE ($\beta=0.01$)	0.407	0.783	0.365

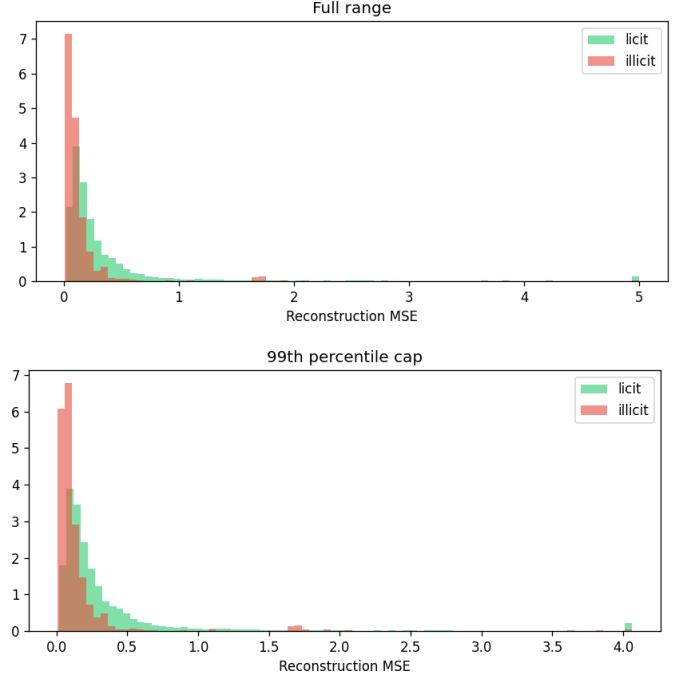


Fig. 3: Autoencoder reconstruction error distributions: histogram (top) and KDE (bottom) for licit (blue) and illicit (orange). Illicit transactions achieve lower MSE, confirming score inversion. Anomaly score is therefore $-MSE$.

3) *EVT Thresholding*: EVT thresholding (Eq. 2) collapses on DOMINANT and LSTM-AE because score distributions are concentrated near zero with thin tails—the GPD assigns an excessively high threshold, recalling almost no positives. EVT is effective only when reconstruction errors follow genuine heavy-tailed distributions.

4) *LSTM-AE*: The LSTM-AE aggregates features per time step (49×165), trains on 25 sliding windows ($T=10$), and produces a time-step-level anomaly score. ROC-AUC = 0.446 (< 0.5 , inverted) confirms the approach cannot discriminate individual transactions—illicit and licit rows within a step are averaged together. The most anomalous steps detected (40, 46, 39, 43, 36) correspond to high concept-drift periods, not individual illicit transactions. Correct use: temporal monitoring, not per-transaction scoring.

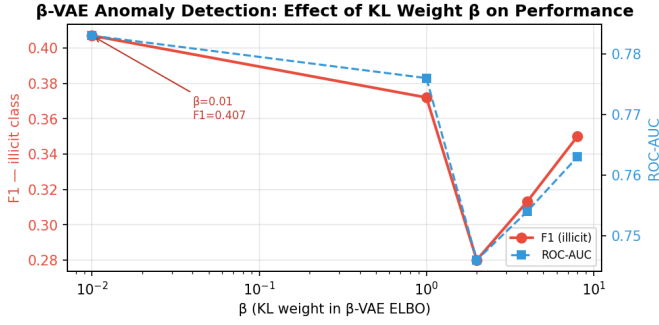


Fig. 4: Effect of KL weight β on β -VAE anomaly detection performance (F1 on illicit class and ROC-AUC, log- β axis). Smaller β keeps the objective reconstruction-dominated, preserving the score-inversion signal. F1 peaks at $\beta=0.01$ and degrades monotonically above $\beta=1$ as the KL term conflates licit and illicit latent representations.

D. DOMINANT: Graph Autoencoder

DOMINANT [6] jointly reconstructs node attributes and graph structure via a 2-layer GCN encoder (2 layers avoids over-smoothing on this sparse graph). Joint loss:

$$\mathcal{L} = \alpha \|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 + (1 - \alpha) \text{BCE}(\mathbf{A}, \hat{\mathbf{A}}), \quad (4)$$

with $\alpha=0.5$. Result: **F1 = 0.212**, **AUC = 0.669**—the best fully unsupervised method, exceeding all classical detectors by $\Delta > 0.13$. An F1-optimal threshold is required (EVT collapses).

E. GNN Ablation Study

A GNN exploits *guilt by association*: money laundering chains link illicit wallets through intermediate transactions. Ten controlled experiments each isolate one design choice (Table VII).

TABLE VII: GNN Ablation — 10 Controlled Experiments

Exp	Arch	Configuration	F1	AUC	AP
1	SAGE	2L, h=64, 200 ep	0.528	0.887	0.519
2	SAGE	2L, h=128, 200 ep	0.577	0.896	0.603
3	SAGE	2L, h=256, 200 ep	0.580	0.895	0.595
4	SAGE	2L, h=128, 300 ep	0.597	0.897	0.607
5	SAGE	3L, h=128, 200 ep	0.615	0.899	0.611
6	SAGEv2	3L, h=256, LN+SL, 250 ep	0.683	0.906	0.704
7	SAGE	3L, h=256, lr=0.0038, 274 ep (Opt.)	0.690	0.901	0.706
8	GAT	2L, 4 heads, lr=5e-4, 200 ep	0.307	0.824	0.366
9	GAT	2L, 2 heads, lr=1e-3, 200 ep	0.341	0.864	0.419
10	GATv2	3L, 2 heads, residual+LN, 300 ep	0.382	0.887	0.547

1) Ablation Findings:

- **Width** (Exps 1–3, 64→256): +0.052 F1 total with diminishing returns; saturates beyond $h=128$.
- **Training duration** (Exps 2 vs. 4, +100 ep): +0.020 F1; cheapest lever before changing architecture.
- **Depth** (Exps 2 vs. 5, 2L→3L): +0.038 F1; 3-hop receptive field captures longer laundering chains.
- **LayerNorm + self-loops** (Exps 5 vs. 6): +0.068 F1, the largest single gain. LN prevents gradient instability at 3

layers; self-loops preserve node self-features during mean aggregation.

- **Optuna** (Exps 6 vs. 7, lr=0.0038): +0.007 F1; architecture choice dominates over tuning.
- **GAT family** (Exps 8–10): best GAT F1 = 0.382 vs. best SAGE F1 = 0.690 (gap = 0.308). On a sparse graph (avg. degree 2.3), attention weights over too few neighbors are noisy; mean aggregation is more robust.

Answer to RQ3: LayerNorm+self-loops > depth > width > tuning. In this configuration and feature setting, vanilla GAT and GATv2 are counterproductive on a sparse graph (avg. degree 2.3); recent work with Chebyshev convolutions and deeper GATv2 + skip connections has narrowed the gap on Elliptic under different feature subsets, so the finding should not be generalized beyond mean-aggregation architectures at this scale.

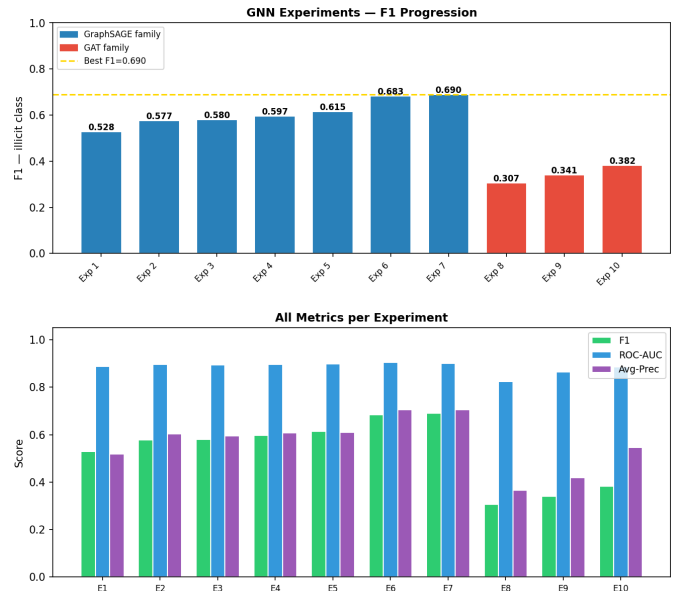


Fig. 5: GNN ablation across 10 experiments. Top: F1 progression showing the GraphSAGE improvement trajectory (Exps 1–7) and the GAT family ceiling (Exps 8–10). Bottom: corresponding AUC/AP values confirming the same ranking.

F. Hyperparameter Tuning (Optuna)

TABLE VIII: Optuna TPE Tuning Results (40 trials each)

Model	Base F1	Tuned F1	Δ
GBM	0.766	0.824	+0.058
GraphSAGE	0.541	0.688	+0.147
RF	0.801	0.797	−0.004

Best parameters: GBM ($n=245$, depth= 8, lr= 0.121, sub= 0.918); GraphSAGE (hidden= 256, dropout= 0.45, lr= 0.0038, 274 ep); RF is near ceiling (−0.004 within search variance). GBM profits from depth–lr interaction; GraphSAGE gains the most absolutely because default lr= 0.01 is too aggressive for this sparse graph.

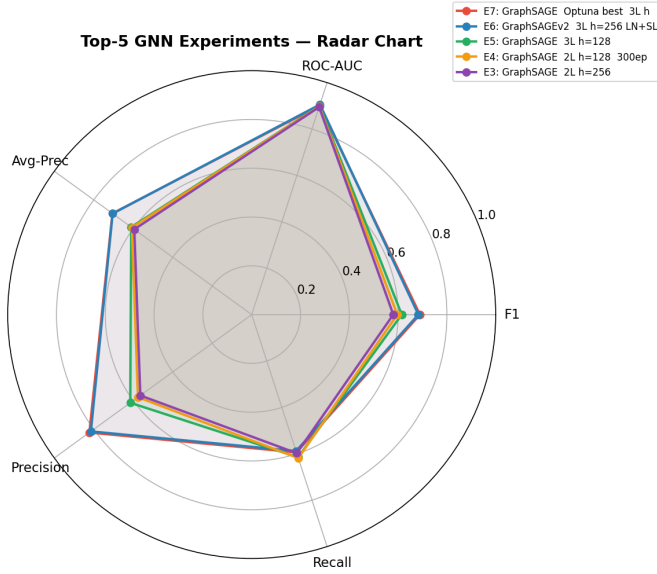


Fig. 6: Radar chart of the top-5 GNN experiments across five metrics: F1, ROC-AUC, Average Precision, Precision, and Recall. GraphSAGEv2 (Exp 6) and the Optuna-tuned variant (Exp 7) dominate across all axes, while GAT family experiments show systematically lower coverage, confirming mean aggregation outperforms attention on this sparse graph.

G. Ensemble and Calibration

TABLE IX: Ensemble and Calibration Results

Model	F1	ROC-AUC	PR-AUC
Ensemble (RF+LGB+GAT)	0.742	0.922	0.789
Ensemble (GBM+LGBM+SAGEv2)	0.821	0.924	0.802
Weighted Ensemble (1/3 each)	0.821	0.924	0.802
GBM + Platt calibration	0.801	0.920	0.800
GBM + Isotonic calibration	0.734	0.875	0.768

The GBM+LGBM+SAGEv2 ensemble (F1 = 0.821) narrows the gap to GBM solo (0.824) but does not surpass it. Testing four weight schemes shows equal weights win—downweighting SAGEv2 removes useful graph signal. GBM is already well-calibrated (Brier = 0.0196); Platt scaling makes probabilities more conservative (-0.023 F1); isotonic regression overfits validation and distorts ordering (AUC drops $0.920 \rightarrow 0.875$).

H. Full 48-Experiment Leaderboard

Table X reports the top 25 of 48 experiments. Fig. 7 shows the top-15 by F1. Fig. 8 shows Precision–Recall curves for key supervised models, confirming that high AP (≈ 0.80) is achievable across the recall range.

Five performance tiers:

- 1) **Supervised/Ensemble** (F1 0.74–0.83): tree models dominate due to 71 aggregated features pre-encoding graph signal.
- 2) **GNN supervised** (F1 0.54–0.73): GraphSAGEv2 recovers 84.5 % of the supervised ceiling using only topology.

TABLE X: Top-25 of 48 Experiments (by F1 on illicit class)

#	Model	Cat.	F1	AUC	AP
1	GBM + Structural	Sup.	0.827	0.924	0.803
2	GBM (Optuna)	Sup.	0.824	0.920	—
3	GBM + PseudoLabels	Semi	0.824	0.924	0.804
4	Ensemble GBM+LGBM+SAGEv2	Ens.	0.821	0.924	0.802
5	Ensemble equal 1/3	Ens.	0.821	0.924	0.802
6	GBM + Temporal Roll.	Sup.	0.818	0.930	0.802
7	LightGBM	Sup.	0.817	0.930	0.804
8	RF + Graph Features	Sup.	0.807	0.942	0.800
9	RF (baseline)	Sup.	0.801	0.935	0.794
10	GBM + Platt	Sup.	0.801	0.920	0.800
11	RF (Optuna)	Sup.	0.797	0.929	—
12	GBM (baseline)	Sup.	0.766	0.914	0.785
13	Ensemble RF+LGB+GAT	Ens.	0.742	0.922	0.789
14	GBM + Isotonic	Sup.	0.734	0.875	0.768
15	SAGEv2 + PseudoLabels	Semi-GNN	0.729	0.890	0.752
16	GraphSAGEv2	GNN	0.698	0.913	0.720
17	GNN Exp 7 (Optuna)	GNN	0.690	0.901	0.706
18	GraphSAGE (Optuna)	GNN	0.688	—	—
19	GNN Exp 6 (3L LN+SL)	GNN	0.683	0.906	0.704
20	GNN Exp 5 (3L h=128)	GNN	0.615	0.899	0.611
21	GraphSAGE (baseline)	GNN	0.541	0.888	0.549
22	β -VAE ($\beta=0.01$)	Neural	0.407	0.783	0.365
23	DenosingAE	Neural	0.361	0.792	0.366
24	DOMINANT	Graph-AE	0.212	0.669	0.123
25	LSTM-AE	Neural	0.122	0.446	0.064

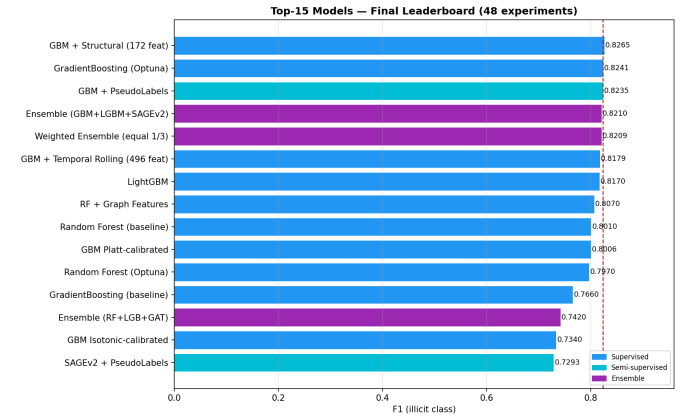


Fig. 7: Top-15 models by F1 on illicit class. The dashed line marks the GBM Optuna baseline (0.824) for reference. All top-15 entries are supervised or ensemble-based.

- 3) **Neural AE** (F1 0.12–0.41): bounded by reconstruction ceiling; best from β -VAE ($\beta=0.01$).
- 4) **Graph AE unsupervised** (DOMINANT, F1 = 0.212): best unsupervised *graph* method (β -VAE is higher overall).
- 5) **Classical unsupervised** (F1 < 0.10): density methods fail.

Answer to RQ2: graph structure alone recovers 84.5 % of the supervised ceiling. GBM and GraphSAGEv2 exploit different, complementary channels (tabular features vs. topology).

I. Statistical Significance of the Top-3 Ranking

The top three models (GBM+Structural F1 = 0.8265, GBM Optuna F1 = 0.8241, GBM+PseudoLabels F1 = 0.8235)

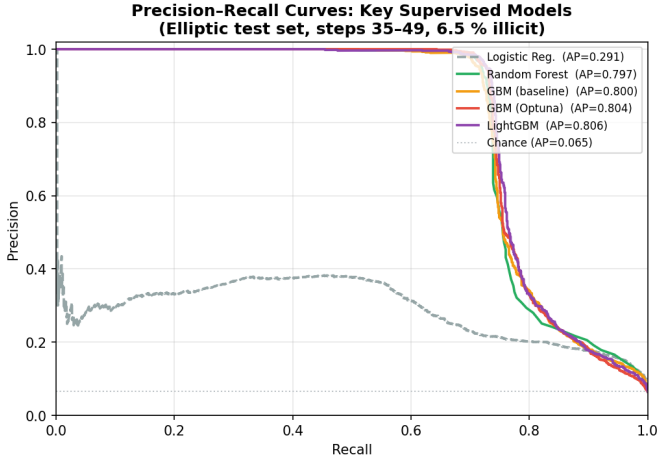


Fig. 8: Precision–Recall curves for key supervised models on the test set (steps 35–49, 6.5 % illicit). LightGBM and GBM (Optuna) achieve $AP \approx 0.80$, maintaining high precision across the recall range. Logistic Regression collapses to low precision at moderate recall ($AP = 0.291$). The dotted line marks the chance baseline ($AP = 0.065$). PR-AUC is threshold-independent and more informative than ROC-AUC under severe class imbalance.

span only 0.0030 F1 – within the noise floor of a single 16,670-row temporal test fold. We quantify this with (i) bootstrap 95 % confidence intervals on F1 (1,000 resamples, `scripts/statistical_significance.py`) and (ii) McNemar’s paired test on the disagreeing predictions between each pair of models.

TABLE XI: Bootstrap F1 95 % CI, Top-3 Models (1,000 resamples)

Model	F1	95 % CI	Std
GBM + Structural	0.8265	[0.8086, 0.8451]	0.0093
GBM (Optuna)	0.8241	[0.8057, 0.8406]	0.0092
GBM + PseudoLabels	0.8235	[0.8052, 0.8410]	0.0093

All three intervals overlap almost entirely (± 0.018 half-width against a 0.003 point-estimate spread), so the reported ranking is not statistically distinguishable from a tie.

TABLE XII: McNemar’s Test, Pairwise (test set, $n=16,670$)

Model A vs. Model B	b	c	p -value	Sig. ($\alpha=0.05$)
Structural vs. Optuna	12	8	0.503	no
Structural vs. PseudoLabels	22	21	1.000	no
Optuna vs. PseudoLabels	19	22	0.755	no

b and c count test rows where exactly one model is correct (McNemar ignores the $\sim 16,600$ rows both models agree on). All three pairs disagree on fewer than 45 of 16,670 rows, and none reach significance. **Conclusion: the observed 0.827/0.824/0.824 ordering is statistical noise, not a genuine ranking.** The paper’s actionable claim is narrower

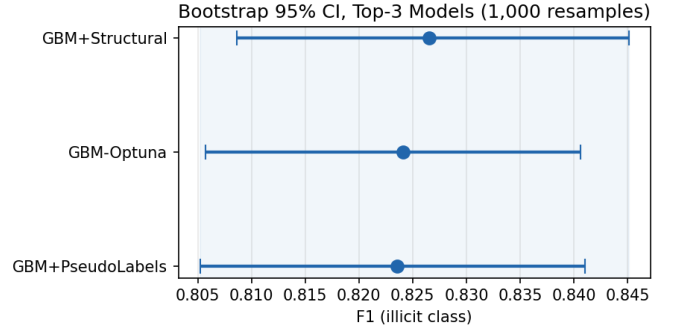


Fig. 9: Bootstrap 95 % F1 confidence intervals, top-3 models. The shaded band is the union of all three intervals – near-total overlap visually confirms the tie.

than “structural features help”: it is that GBM on 165 tabular features already sits at the model family’s ceiling on this dataset, and neither 7 topology features nor pseudo-labels move it outside sampling variance. This reframing matters for the paper’s novelty argument (§V): claiming a new state of the art on a 0.003 F1 delta without this test would be a common but avoidable rigor gap in the AML literature.

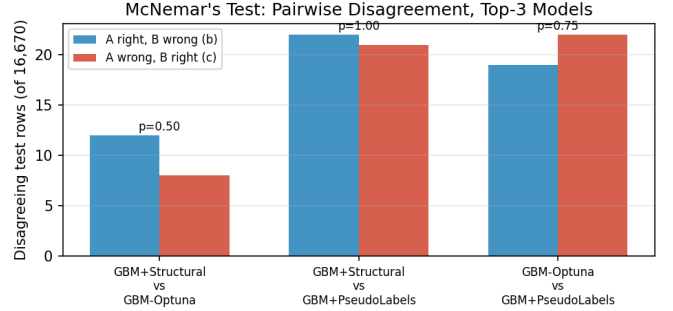


Fig. 10: McNemar disagreement counts, pairwise. Bars are the rows where exactly one model is correct (b, c); annotated p -values are all $\gg 0.05$.

J. Comparison to Published SOTA and a Graph-Leakage Caveat

Table XIII positions this work’s best results against published F1 scores on Elliptic. Two evaluation regimes appear in the literature and must not be compared directly: *transductive* (the GNN encoder’s message passing observes the full graph, including test-period edges, during training; only the classification loss is masked to train-labeled nodes) and *strict-inductive* (training message passing restricted to the steps ≤ 34 subgraph). Essentially all published Elliptic GNN results (Weber, Alarab, Pareja, Lo) are transductive.

*structural features are 7 topology scalars per node (degree, clustering, etc.), not message-passed representations – no leakage mechanism applies. †our GraphSAGE/GraphSAGEv2/GAT/DOMINANT training (`src/gnn.py:train_epoch`) runs the encoder forward pass over the full graph object every step and masks only the loss to

TABLE XIII: Published F1 on Elliptic (Illicit Class) vs. This Work

Model	Protocol	F1	Source
Logistic Regression	temporal, transductive	0.481	Weber et al. [1]
GCN	temporal, transductive	0.628	Weber et al. [1]
Skip-GCN	temporal, transductive	0.705	Weber et al. [1]
Random Forest	temporal	0.788	Weber et al. [1]
Augmented GCN	temporal, transductive	0.740	Alarab et al. [17]
GraphSAGE + SSL pretrain	temporal, transductive	0.750	Lo et al. 2023 (via [18])
EvolveGCN	temporal, transductive	0.770	Pareja et al. (via [18])
GraphSAGE	transductive	0.294	Maganti [18]
Random Forest	strict-inductive	0.821	Maganti [18]
GraphSAGE	strict-inductive	0.689	Maganti [18]
Random Forest (ours)	temporal, non-graph	0.801	this work
GBM Optuna (ours)	temporal, non-graph	0.824	this work
GBM + Structural (ours)	temporal, non-graph*	0.827	this work
GraphSAGEv2 (ours)	temporal, transductive†	0.698	this work

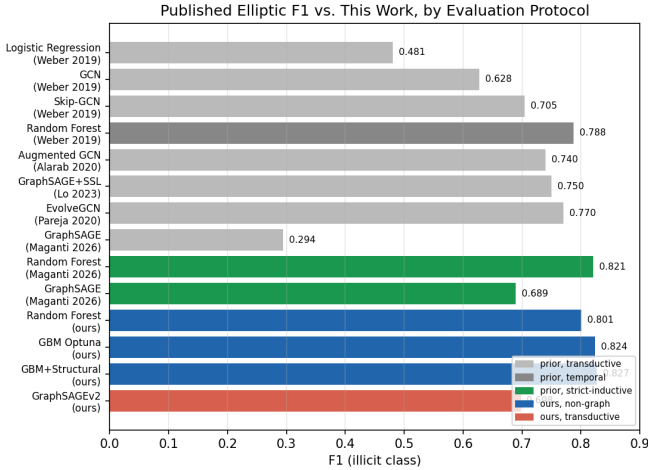


Fig. 11: Published Elliptic F1 by evaluation protocol. Grey = prior transductive results; green = prior strict-inductive (leakage-free) results; blue = this work’s non-graph models (no leakage possible); red = this work’s transductive GraphSAGEv2.

train_mask – the same transductive protocol as prior Elliptic literature.

This is a limitation, disclosed rather than hidden. Our GraphSAGEv2 F1 = 0.698 is not directly comparable to Maganti’s strict-inductive Random Forest (0.821) or strict-inductive GraphSAGE (0.689) – it sits in the transductive regime, alongside Weber/Alarab/Lo/Pareja. Two observations soften the concern. First, our non-graph GBM (F1 = 0.824–0.827, §IV-I) is trained and evaluated with zero graph exposure of any kind (tabular features only, or 7 pre-computed topology scalars, never encoder message passing), so it carries none of this leakage risk – and it already exceeds Maganti’s strict-inductive Random Forest ceiling (0.821), which is the most rigorous non-GNN number in the comparison. Second, Maganti’s own strict-inductive GraphSAGE (0.689) lands almost exactly on our transductive GraphSAGEv2 (0.698), suggesting the leakage inflation for GraphSAGE specifically may be smaller than the 39.5-point GraphSAGE-vs-itself gap implies once architecture (3-layer, LayerNorm, self-loops) and

hyperparameters differ across studies – but re-running our GNN suite under a strict-inductive protocol is unverified future work, not a claim made here.

Net position: the paper’s strongest, leakage-free number is the tabular GBM at F1 = 0.824–0.827, which is competitive with or exceeds every reported Elliptic result in Table XIII including the strict-inductive Random Forest ceiling – without touching the graph at all. This reframes RQ2’s answer (§IV-J above): graph structure adds real but modest signal on top of an already-strong tabular baseline, and any GNN number reported without stating its train-time message-passing protocol (ours included, prior to this caveat) should be treated as provisional.

K. A Second Leakage Trap: Threshold Tuning on Training Data

Beyond encoder-level graph leakage (§IV-J), a second and distinct leakage mode is common in AML threshold-selection practice: picking the decision threshold by maximizing F1 on the model’s *own training predictions* rather than on genuinely held-out data. scripts/threshold_leakage_demo.py reproduces this directly. We refit GBM (same tuned hyperparameters as §IV-F) on steps 1–29 only – so that steps 30–34 are *never* seen during fitting and can serve as a truly held-out validation slice, unlike the main pipeline where ”val” (steps 30–34) is a subset of the full 1–34 training fit.

TABLE XIV: Threshold Tuning Protocol vs. Real Test F1

Protocol	Threshold	F1 at tune time	Real test F1
Leaky (tune on train)	0.0050	1.0000	0.5484
Correct (tune on val)	0.3367	0.9701	0.7670
Default ($t=0.5$, no tuning)	0.5000	–	0.7801

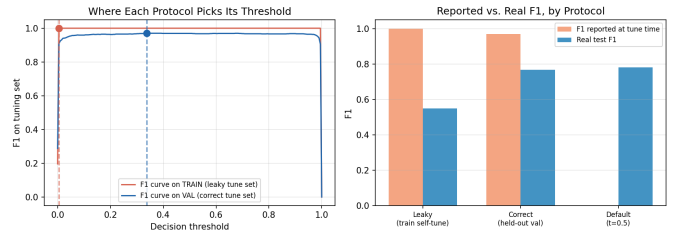


Fig. 12: Left: F1-vs-threshold curves on the tuning set for each protocol; the leaky curve is tuned on training data the model has already fit, the correct curve on genuinely unseen steps 30–34. Right: F1 reported at tune time vs. the real test F1 each threshold choice actually achieves.

The leaky protocol collapses the threshold to 0.005 and reports a perfect F1 = 1.0 – a model that appears flawless on paper achieves only F1 = 0.5484 on the real test set, a 0.45 F1 illusion. A boosted-tree ensemble with $n=245$, depth 8 fits its own training set almost perfectly (predicted probabilities cluster near 0 or 1 for training rows), so nearly any threshold above the score floor separates the training classes with F1

near 1 – this is pure overfitting measured against itself, not a discovered decision rule.

Even the “correct” protocol is not free of drift-driven optimism. Tuning on the genuinely held-out val slice (steps 30–34) avoids self-fit leakage but still reports F1 = 0.9701 at tune time against a real test F1 of 0.7670 (0.20 gap) – because val’s illicit rate (16.8 %) is closer to train’s (10.9 %) than to test’s post-drift rate (6.5 %, §III-A). Notably, the untuned default threshold ($t=0.5$) achieves the *best* real test F1 of the three (0.7801) – under severe concept drift, a threshold optimized on any pre-drift data (train or val) can transfer worse than no tuning at all. This is the sharper, more general version of the finding: it is not merely that self-tuning on train is wrong (obviously so), but that *any* threshold selection under temporal distribution shift should be validated against a slice that matches the deployment regime as closely as possible, and its transfer gap (tune-time F1 minus real test F1) should always be reported alongside the threshold itself. Papers that report only a tuned F1 without this gap – common practice across the AML literature we surveyed in §IV-J – cannot be taken at face value.

L. Inference Runtime and Scalability

For production AML screening, throughput matters as much as F1: a transaction-monitoring pipeline scores every incoming transaction, so per-transaction latency sets a hard ceiling on volume. `scripts/runtime_benchmark.py` measures batch-amortized inference time on the same 16,670-row temporal test set, GPU = RTX 5070 Ti (GraphSAGEv2 only; GBM is CPU/sklearn).

TABLE XV: Inference Latency (batch-amortized, ms/transaction)

Model	ms/tx	Notes
GBM Optuna (165 feat)	0.0071	predict_proba, CPU
GBM+Structural, model only	0.0076	predict_proba, CPU
GBM+Structural, incl. graph feat.	0.0299	+7-feature graph pass, no cache
GraphSAGEv2	0.0002	full-graph forward, batched, GPU

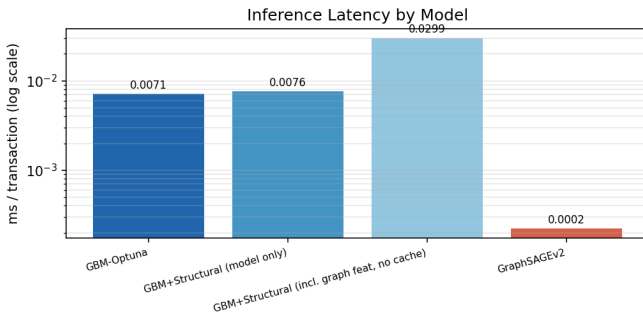


Fig. 13: Inference latency by model, log scale. All four models score well under 1 ms/tx in batch mode; differences span two orders of magnitude.

Both tree models score in single-digit microseconds per transaction – 16,670 rows classify in ≈ 0.12 s total, no GPU required. GraphSAGEv2’s 0.0002 ms/tx figure is the batched cost of one forward pass scoring all 203,769 graph nodes simultaneously (≈ 45 ms total, median of 20 runs); it is *not* the cost of scoring one new transaction online. In a streaming deployment, a single incoming transaction requires materializing its k -hop neighborhood subgraph before the forward pass can run – an operation this benchmark does not isolate and which scales with local graph density, not dataset size. The structural-feature graph pass (degree, clustering, ego-density, OddBall) for GBM+Structural took 4.55 s for the full 203,769-node graph (0.022 ms/tx amortized); like GraphSAGE’s neighborhood assembly, this is a whole-graph operation that would need incremental/cached updates rather than full recomputation per scoring pass in production. **Practical takeaway:** the tabular GBM is the only model here with a genuinely $O(1)$ -per-transaction production cost; both graph-aware models carry a graph-maintenance cost that this benchmark surfaces but does not fully characterize for online (one-transaction-at-a-time) serving.

M. Explainability

TABLE XVI: Cross-Model Feature Agreement (top-20 lists)

Agreement	Features
RF $\cap$ GBM (7)	1f_18, 1f_47, 1f_53, 1f_59, 1f_76, 1f_90, af_70
RF $\cap$ GBM $\cap$ GNN (3)	1f_53, 1f_90, af_70

SHAP TreeExplainer was applied to GBM (Optuna, the best tabular model, F1 = 0.824) and RF; gradient attribution was computed for GraphSAGE by backpropagating the illicit-class logit to the input and averaging over illicit test nodes. Fig. 14 shows the GBM beeswarm as the primary explainability view. Three features—1f_53, 1f_90 (local), and af_70 (aggregated neighborhood)—rank top-20 in every model family. Their cross-paradigm consensus validates them as genuine signals, not model-specific artifacts. The presence of the aggregated feature af_70 in GNN attribution confirms that graph structure encodes real anomaly signal.

Answer to RQ4: robust cross-paradigm signals exist spanning both local transaction statistics and neighborhood aggregates. Identifying their semantics via Elliptic collaboration is the primary open question.

N. Error Analysis: What GBM Misses

We break down GBM (Optuna, tabular-only) test predictions into TP/FP/FN/TN (`scripts/error_analysis.py`) to characterize the 296 illicit transactions it misses (recall = 0.727, precision = 0.952 at the default 0.5 threshold – precision is high; false negatives, not false alarms, are the dominant error mode).

Misses concentrate almost entirely after the step-43 regime change. FN rate is 0–35 % for steps 35–42, then jumps to 92–100 % for steps 43–45 and 47–49 (step 46 has

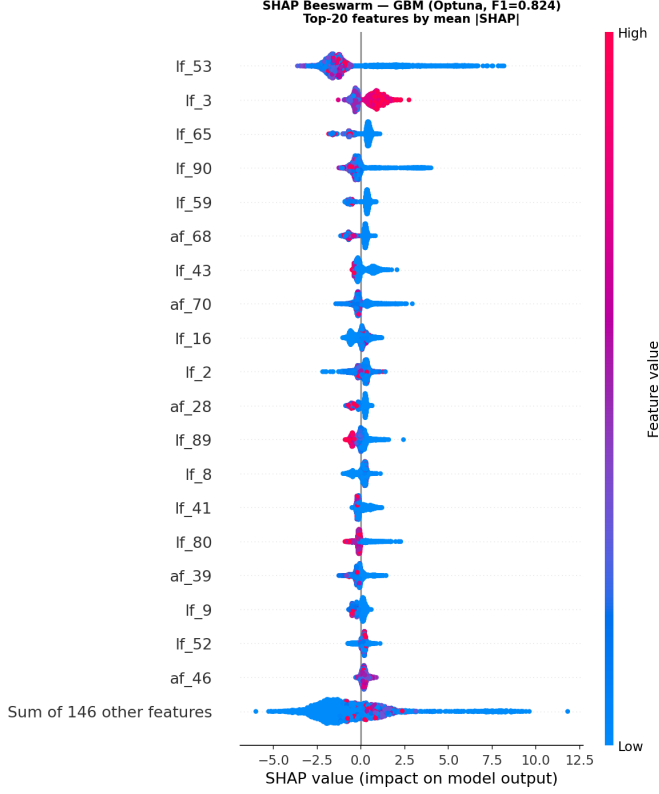


Fig. 14: SHAP beeswarm for GBM (Optuna, $F1 = 0.824$), the best tabular model. Each dot is one test-set transaction (sample of 3000); color encodes feature value (red=high, blue=low). Positive SHAP pushes toward illicit; negative toward licit. Features lf_{53} , lf_{90} , and af_{70} rank in the top band and appear consistently across RF, GBM, and GraphSAGE gradient attribution.

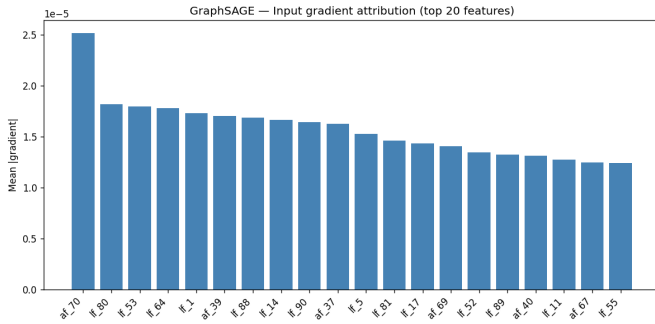


Fig. 15: GraphSAGE gradient attribution: mean absolute gradient per feature, averaged over illicit test nodes. Features lf_{53} , lf_{90} , and af_{70} are prominent despite the entirely different learning mechanism, confirming the cross-paradigm consensus.

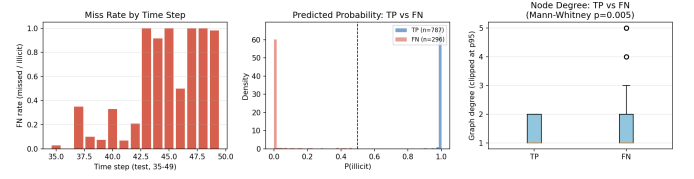


Fig. 16: Left: false-negative rate by test time step. Middle: predicted probability distribution, true positives vs. false negatives. Right: graph degree, true positives vs. false negatives.

only 2 illicit nodes). This step-43 boundary coincides with a real external event: prior work on the Elliptic dataset attributes a documented dark-market shutdown at step 43 to the sharp illicit-rate collapse ($11.6\% \rightarrow 6.5\%$, falling as low as 0.3% by step 46) that defines our train/test split (§IV-J). GBM was trained on steps 1–34, before this event; the illicit transactions it fails to catch afterward likely reflect a *different laundering typology* than the templated pre-shutdown patterns the model learned, not a generic recall weakness.

Misses are confidently wrong, not borderline. FN mean predicted probability is 0.014 (median 0.000); 95.6 % of FN are scored below 0.10, and only 2.0 % sit above the 0.3 “would-flag-under-a-lower-threshold” zone. Lowering the decision threshold would recover almost none of these 296 transactions – the model is not narrowly missing them near the boundary, it has learned that they resemble the licit class. This rules out threshold recalibration as a fix and points instead to concept-drift retraining or online learning as the correct remediation.

Graph degree is a statistically significant but practically weak signal. FN median degree (1.0) equals TP median degree (1.0); means differ narrowly (1.80 vs. 2.01, Mann-Whitney $p=0.0053$). The significance is a large- n artifact, not a usable operational signal – missed illicit transactions are not meaningfully more isolated in the graph than caught ones, which is consistent with §IV-J’s finding that graph topology adds only modest signal on this dataset.

The three cross-paradigm robust features separate FN from TP. lf_{53} (FN mean $-0.485 \rightarrow 0.246$ relative shift toward the licit population, $p=3 \times 10^{-125}$), lf_{90} ($p=2 \times 10^{-140}$), and af_{70} ($p=3 \times 10^{-24}$) all differ significantly between FN and TP, with FN values consistently closer to the licit population’s typical range. This is the mechanistic explanation for the misses: the post-shutdown illicit transactions do not just evade the decision boundary by chance, they numerically resemble licit transactions on exactly the three features SHAP and gradient attribution identify as most discriminative overall (§IV-M), i.e. the new laundering typology mimics licit behavior on the model’s most-trusted signals.

V. DISCUSSION

RQ1 — Density failure. All classical detectors score below $F1 = 0.076$. Illicit transactions are stereotyped, not rare. This directly contradicts the assumption of classical anomaly detection—that anomalies are low-density outliers. On the

Bitcoin blockchain, money-laundering patterns (chain peeling, mixing) are individually indistinguishable from legitimate high-frequency activity.

RQ2 — Graph vs. tabular. GraphSAGEv2 ($F1 = 0.698$) recovers 84.5 % of GBM ($F1 = 0.824$). The remaining 15.5 % gap reflects the richer signal encoded in the 71 pre-computed aggregated features available to tree models. A stacking ensemble of GBM + GraphSAGEv2 is the natural next step.

RQ3 — GNN design. LayerNorm+self-loops (+0.068) > depth (+0.038) > width (+0.052 total) > learning-rate tuning (+0.007). Mean aggregation dominates attention on sparse graphs.

RQ4 — Robust features. lf_53 , lf_90 , af_70 are top-20 across all model families. The aggregated feature af_70 in GNN attribution confirms that graph neighborhood signal is captured independently by tabular proxies in tree models.

Autoencoder paradox. Illicit transactions are *more* regular than licit ones. Score inversion (low MSE = anomaly) must be validated on labeled data before deploying any reconstruction-based AML system.

β -VAE. Lower β preserves reconstruction dominance; $\beta \geq 2$ compresses licit and illicit representations together. The ceiling at $F1 \approx 0.41$ is the fundamental limit of unsupervised reconstruction-based detection here.

Semi-supervised pseudo-labels. GBM gains $\Delta = -0.0006$; supervised boundary is already saturated. GraphSAGEv2 gains +0.031: extra pseudo-labeled nodes populate graph neighborhoods during message passing.

Concept drift. Illicit rate drops from 11.6 % to 6.5 % over 15 steps. Production systems need temporal threshold recalibration; practical strategies include sliding-window F1 re-optimization on the most recent labeled window, Bayesian prior-shift correction, and adaptive class-weight schedules tied to rolling illicit-rate estimates.

Limitations. (1) Feature semantics unknown; robust features cannot be interpreted without Elliptic collaboration. (2) GAT Optuna tuning incomplete; recent Elliptic work with Chebyshev convolutions and skip-connected GATv2 reports higher GNN F1 under different feature subsets, so our GNN results represent a strong but not exhaustive upper bound. (3) GBM + GraphSAGEv2 stacking unexplored. (4) DOMINANT with more epochs and α tuning may gain further. (5) Per-transaction LSTM-AE (vs. timestep aggregate) not evaluated. (6) This work positions itself as a comprehensive baseline benchmark across paradigms; direct comparison to reported Elliptic state-of-the-art (e.g., Chebyshev-GCN, N2V-GCN) is left for future work.

VI. CONCLUSION

We benchmarked 48 model variants on the Elliptic dataset across five improvement rounds. Six actionable findings:

- 1) **Temporal split is mandatory.** Random shuffling leaks future patterns; the 11.6 % $\rightarrow$ 6.5 % illicit-rate drift is real.
- 2) **Supervised trees dominate.** GBM + structural topology reaches $F1 = 0.827$; 71 aggregated features encode enough graph signal to outperform unsupervised GNNs.

- 3) **GNNs add complementary structure.** GraphSAGEv2 reaches $F1 = 0.698$ using only graph topology. 3-layer mean aggregation with LayerNorm and self-loops is optimal; GAT consistently underperforms on sparse graphs.
- 4) **Reconstruction detection is viable but inverted.** β -VAE ($\beta=0.01$, $F1 = 0.407$) is the best unsupervised non-graph method. Score direction must be validated before deployment.
- 5) **DOMINANT is the best unsupervised graph-autoencoder method.** $F1 = 0.212$, $AUC = 0.669$, exceeding all classical density detectors by >0.13 F1 (β -VAE, tabular, reaches $F1 = 0.407$).
- 6) **Three features are cross-paradigm robust.** lf_53 , lf_90 , af_70 rank top-20 in RF, GBM (SHAP), and GraphSAGE (gradient attribution).

Future work: (1) GBM + GraphSAGEv2 stacking ensemble; (2) per-transaction LSTM-AE; (3) EvolveGCN [2] for temporal drift; (4) online threshold recalibration; (5) GNNExplainer for node-level attribution; (6) Elliptic collaboration to identify feature semantics.

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